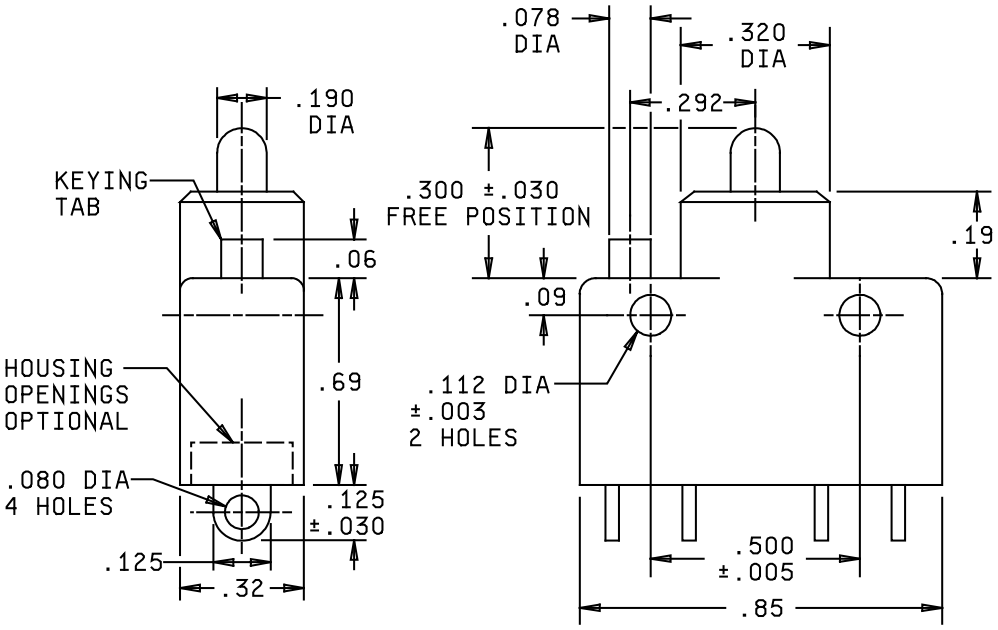
INCH-POUND
MIL-PRF-8805/84F
6 December 2012
SUPERSEDING
MIL-PRF-8805/84E
4 October 2004

PERFORMANCE SPECIFICATION SHEET

# SWITCHES, SENSITIVE, WATERTIGHT, 10 AMPERES, DOUBLE BREAK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the switches described herein shall consist of this specification and the latest issue of MIL-PRF-8805.

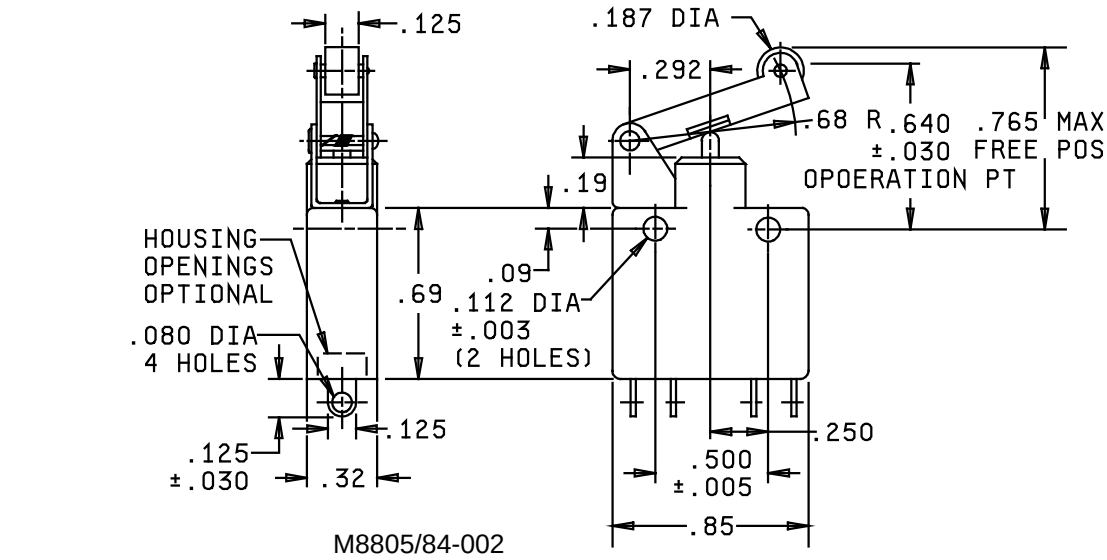



M8805/84-001

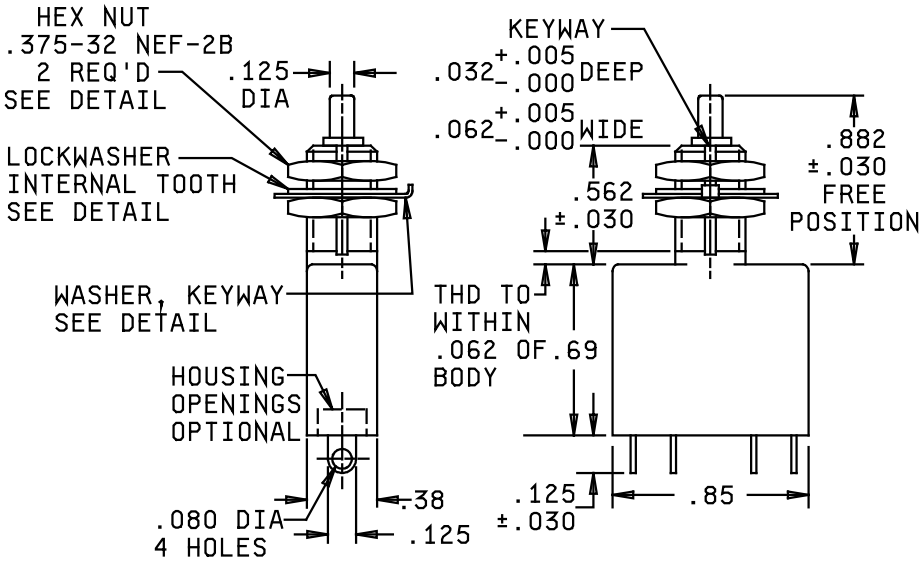
FIGURE 1. Dimensions and configurations.



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M8805/84-002

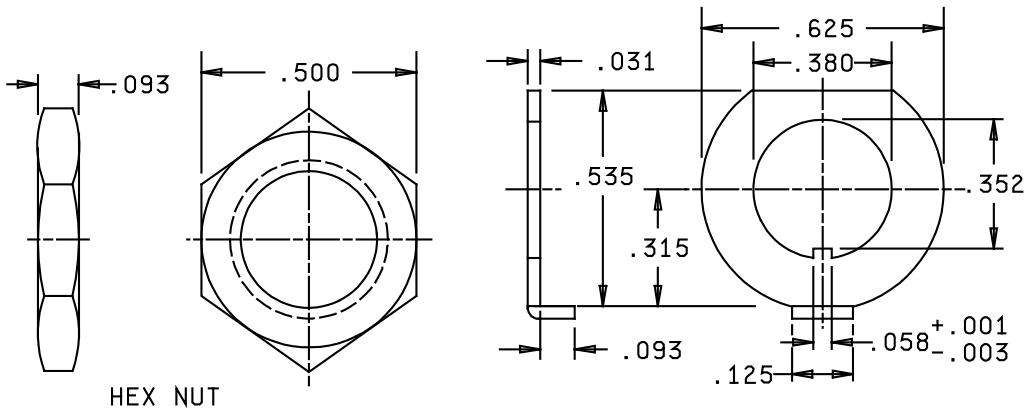



M8805/84-003

FIGURE 1. Dimensions and configurations - Continued.



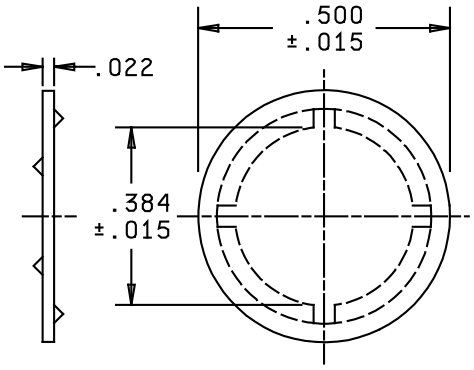
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HEX NUT

Can be replaced by
MS25082-C20

KEYWAY WASHER




INTERNAL TOOTH
LOCKWASHER

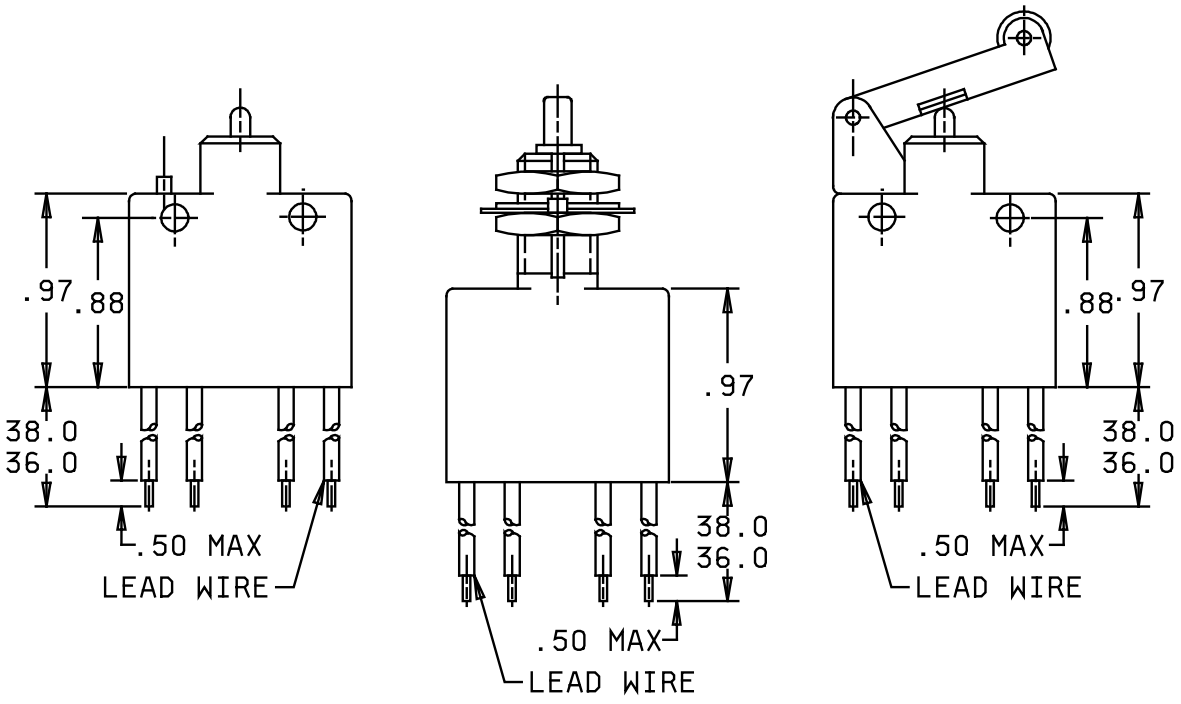
Can be replaced by NASM 35333-134

M8805/84-003

FIGURE 1. Dimensions and configurations - Continued.



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M8805/84-004
(same as -001
except as shown)

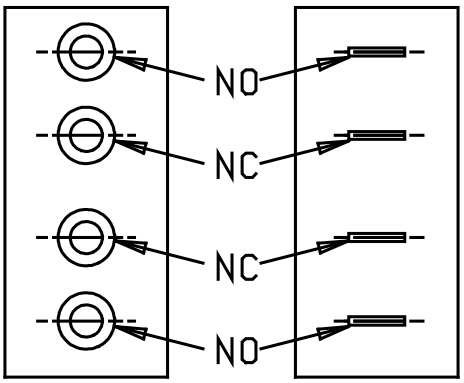
M8805/84-006
(same as -003
except as shown)

M8805/84-005
(same as -002
except as shown)

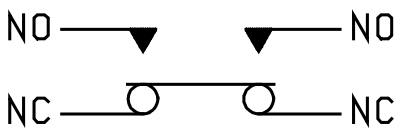
FIGURE 1. Dimensions and configurations - Continued.



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TERMINAL IDENTIFICATION




CIRCUIT SCHEMATIC

| Inches | mm | Inches | mm | Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|---|---|---|---|
| .002 | 0.05 | .06 | 1.5 | .125 | 3.18 | .352 | 8.94 | .68 | 17.3 |
| .003 | 0.08 | .062 | 1.57 | .187 | 4.75 | .375 | 9.53 | .69 | 17.5 |
| .005 | 0.13 | .068 | 1.73 | .19 | 4.8 | .380 | 9.65 | .765 | 19.43 |
| .010 | 0.25 | .078 | 1.98 | .250 | 6.35 | .384 | 9.75 | .85 | 21.6 |
| .015 | 0.38 | .080 | 2.03 | .292 | 7.42 | .500 | 12.70 | .88 | 22.4 |
| .02 | 0.5 | .09 | 2.3 | .300 | 7.62 | .535 | 13.59 | .882 | 22.40 |
| .022 | 0.56 | .093 | 2.36 | .315 | 8.00 | .562 | 14.27 | .97 | 24.6 |
| .025 | 0.64 | .109 | 2.77 | .32 | 8.1 | .625 | 15.88 | 36.00 | 914.4 |
| .030 | 0.76 | .112 | 2.84 | .320 | 8.13 | .640 | 16.26 | 38.00 | 965.2 |
| .031 | 0.79 | .120 | 3.05 | | | | | | |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.02 (0.5 mm) on two place decimals and ±.010 (0.25 mm) on three place decimals.
4. Terminal designation:
   M8805/84-001, M8805/84-002, M8805/84-003: NO and NC stamped on bottom or either side of switch. M8805/84-004, M8805/84-005, M8805/84-006: NO-20 and NC-20 stamped on leads. The marking shall be legible, permanent, shall not impair the wire characteristics, shall appear every 15 inches (381. mm) maximum, and shall appear within three inches (76.2 mm) of the end.
5. Switch design optional within envelope dimensions shown.

FIGURE 1. Dimensions and configurations - Continued.



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REQUIREMENTS:

Dimensions and configurations: See figure 1.

Enclosure design: 3 (watertight).

Temperature characteristic: 1 (-55°C to +85°C).

Shock type: M (100 g's, test condition I, method 213 of MIL-STD-202).

Vibration grade: 1 (10 Hz to 500 Hz swept sinusoidal, 10 g's peak).

Finish: Switch housing shall be processed to resist corrosion.

Weight:

M8805/84-001 - .50 ounce maximum.
M8805/84-002 - .60 ounce maximum.
M8805/84-003 – 1.10 ounce maximum.
M8805/84-004 – 2.30 ounce maximum.
M8805/84-005 – 2.40 ounce maximum.
M8805/84-006 – 2.90 ounce maximum.

Operating characteristics: See table I.

| Contact resistance (maximum): | M8805/84-001, -002, -003 | M8805/84-004, -005, -006 |
|---|---|---|
| Initial | 50 milliohms. | 150 milliohms. |
| After mechanical endurance - | 100 milliohms. | 200 milliohms. |
| Terminal strength: | | |
| Force (Static) - | 5 lbs. | 5 lbs. |
| Duration of force - | 5-10 seconds. | 5-10 seconds. |

Mechanical endurance: 50,000 cycles.

Electrical endurance: 50,000 cycles.

Electrical ratings: See table II.

Intermediate current: 50,000 cycles.

Part or Identifying Number (PIN): See table I.

Qualification:

Group submission – See table III.



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TABLE I. PIN and characteristics. 1/

| PIN M8805/84 | Type of actuator | Actuating force (maximum) | Releasing force (minimum) | Pretravel (maximum | Movement differential | Overtravel (minimum) | Strength of actuator |
|---|---|---|---|---|---|---|---|
| -001<br>-004 | Pin plunger | 2.5 lbs | .5 lb | (inch)<br>.050<br>(1.27 mm) | (inch)<br>.020 ±.010<br>(.51 ±.25 mm) | (inch)<br>.060 max<br>(1.52 mm) | (lbs)<br>25 |
| -002<br>-005 | Roller-lever | 20 oz | 4 oz | .125<br>(3.18 mm) | .060 max<br>(1.52 mm) | .025<br>(.64 mm) | 10 |
| -003<br>-006 | Overtravel plunger | 2.5 lbs | .5 lb | .070<br>(1.78 mm) | .020<br>(.51 ±.25 mm) | .187<br>(4.75 mm) | 25 |

1/ ±20 percent variation in values acceptable after test.

TABLE II. Electrical ratings.

| Load | Sea level - 28 V dc | | 50,000 feet - 28 V dc | |
|---|---|---|---|---|
| | Single circuit NO or NC | 2 circuit 1/ | Single circuit NO or NC | 2 circuit 1/ |
| | (amperes) | (amperes) | (amperes) | (amperes) |
| Resistive | 10 | 10 | 10 | 10 |
| Inductive | 7.5 | 5 | 3.5 | 2 |
| Motor | 4 | 2 | --- | --- |
| Intermediate current | .04 | .04 | .04 | .04 |

1/ Both NO and NC circuits under load during electrical endurance test.

TABLE III. Qualification inspection.

| Examination or test | Basic switch | Other switch samples | Extent of approval |
|---|---|---|---|
| Qualification inspection table of MIL-PRF-8805 | M8805/84-004 (all sample units) | M8805/84-001, M8805/84-002, M8805/84-003, M8805/84-005 M8805/84-006 (2 sample units). Visual and mechanical examination and operating characteristics. M8805/84-002 and -003 strength of actuating means. M8805/84-005 vibration and shock. M8805/84-001 strength of terminals. | All |



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Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where modifications from this revision were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced Documents:
MIL-PRF-8805
MIL-STD-202
MS25082
NASM35333

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5930-2011-123)

Review activities:
Army - AT, AM, AV, MI
Navy - AS, MC, OS, SH
Air Force - 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/ .



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